Cardiovascular Diseases: How Occupation Influences Risk

Cardiovascular Disease Deaths in Mulnomah County, Oregon from 2013-2023

By Nathaniel Segura

Research Topic:

This project examined the relationship between occupational industry and cardiovascular disease deaths in Mulnomah County. The guiding question was as follows:

> Are there occupations with higher levels of risk for cardiovascular disease?

Background

Xia, et al. (2025)

- Higher CV mortality among zero activity and high activity participants (U-shape)
- Higher CV mortality among lower educated workers

Gao et al. (2024)

- Higher death risk for workers who mainly sit during working hours
- Suggests light physical activity during the day to mitigate risk

Cillekens et al. (2022)

- Literature analysis shows conflicting results on the impact of occupational physical activity

Takeaways:

- Extreme and recurrent physical activity may increase CV death risk
- Levels and categorization of physical activity vary, presenting differing results
- We may see higher CV death rates among occupations with both extremely high and extremely low physical activity levels

Methods

Dataset: Multnomah County Vital Records Data

- 65,898 deaths from 2013 - 2023

Process using R Programming:

- Limit to Cardiovascular Disease-caused deaths (ICD-10 codes)
- Limit to 2013-2023 (eleven-year range)
- Employ CDC's API to create NAICS and ACS codes for each death
    - Remove deaths with N/A codes
    - Remove military codes (not in queried census data)
- Query US Census Data: Table C24030 "Sex by Industry"
- Join datasets: remove sex distinction
- Create death rates per 100k population

Methods: Web API

Initial Format:

- Unstandardized / Inconsistent
- ~30,000 Different Labels

	ddoccup	ddind	n
	<chr>	<chr>	<int>
1	-	-	7
2	- Disabled	- Disabled	1
3	- disabled	- disabled	1
4	- infant	- infant	1
5	-Infant	-Infant	1
6	/ Disabled	/ Disabled	1
7	100% Disabled	Disability	1
8	100% Disabled	Disabled	1
9	10th Grade Student	High School	2
10	1st Vice President	Banking	1

19	9th Grade Student	Education
20	9th Grade Student	High School

7	HOMEMAKE	HOMEMAKER	1
8	HOMEMAKER	HOME	3
9	HOMEMAKER	HOME OWNER	1
10	HOMEMAKER	OWN HOME	68
11	HOMEMAKER	OWN RESIDENCE	1
12	HOMEMAKER	NA	1
13	HOMEMAKER/ VOCALIST	HOUSEHOLD	1

API:

- Returned requested industry code
- Almost 24 hours for each run
- Crosswalk utilized to convert codes into the 13 standardized Industry categories

NIOCCS Industry and Occupation Coding Web API

The result is the following JSON:


```
{
  "Industry": [{
      "Code": "8191",
      "Title": "General Medical and Surgical Hospitals, and Specialty (except Psychiatric and Substance Abuse) Hospitals",
      "Probability": 0.999999166
  }],
  "Occupation": [{
      "Code": "3255",
      "Title": "Registered Nurses",
      "Probability": 0.9999963
  }],
  "Scheme": "Census Industry and Occupation 2018"
}
```

Methods: Final Formats

Aggregated Data:

	ddodyear	naicscode	deathcount
1	2013	11	28
2	2013	21	3
3	2013	22	8
4	2013	23	89
5	2013	31	52
6	2013	32	35
7	2013	33	58
8	2013	42	23
9	2013	44	102
10	2013	45	24
11	2013	48	71
12	2013	49	15
13	2013	51	26
14	2013	52	37
15	2013	53	24
16	2013	54	60
17	2013	55	3
18	2013	56	33

Formatted Crosswalk File:

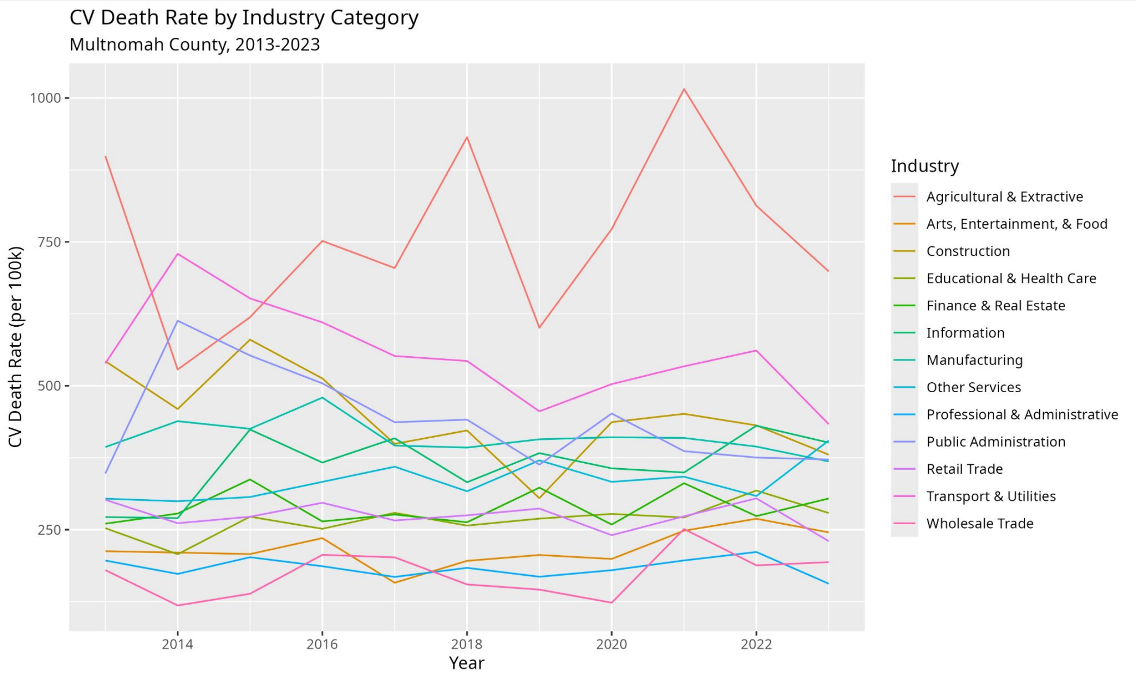
	industrycategory	censusstart	censusend	naicscodestart	naicscodeend
1	Agriculture, Forestry, Fishing and Hunting, and Mining	170	490	11	21
2	Construction	770	770	23	23
3	Manufacturing	1070	3990	31	33
4	Wholesale Trade	4070	4590	42	42
5	Retail Trade	4670	5790	44	45
6	Transportation and Warehousing, and Utilities	6070	6390	48	49
7	Transportation and Warehousing, and Utilities	570	690	22	22
8	Information	6470	6780	51	51
9	Finance and Insurance, and Real Estate and Rental and Leasi...	6870	7190	52	53
10	Professional, Scientific, and Management, and Administrativ...	7270	7790	54	56
11	Educational Services, and Health Care and Social Assistance	7860	8470	61	62
12	Arts, Entertainment, and Recreation, and Accommodation a...	8561	8690	71	72
13	Other Services, Except Public Administration	8770	9290	81	81
14	Public Administration	9370	9590	92	92

Source: United States Census Bureau

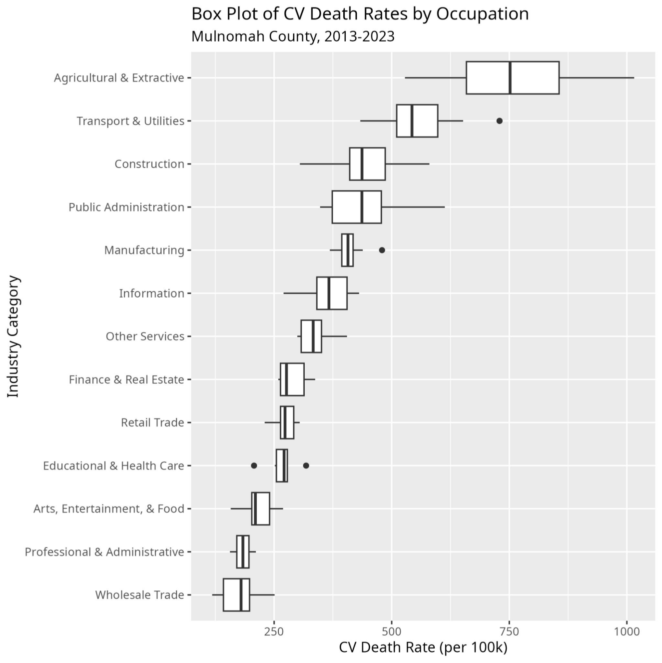
Results:

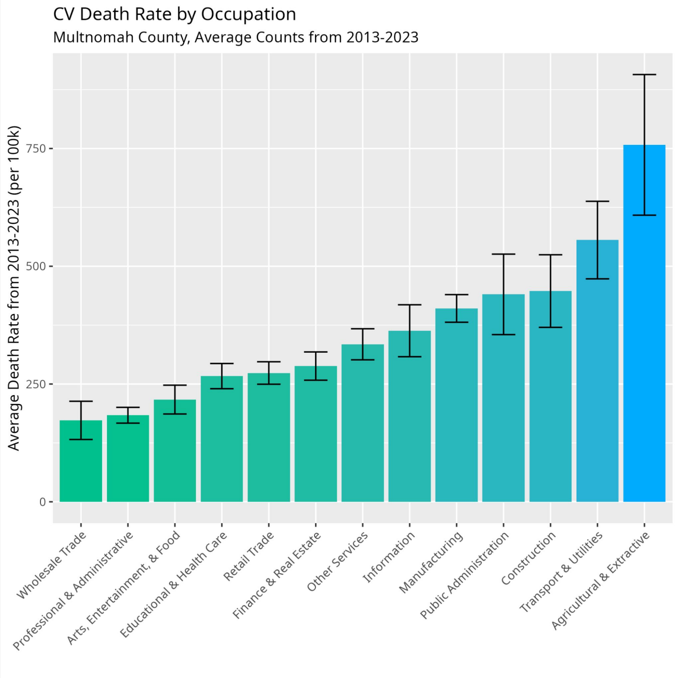
Visualization
s

Visualizations 1



Visualizations (Cont.)





Results: Statistics

This is a statistical analysis of the industries with the 3 highest and 3 lowest average death rates.

Highest:
- Agricultultural & Extractive
- Construction
- Transportation

Lowest:
- Arts, Entertainment, & Food
- Wholesale Trade
- Professional & Administrative

ANOVA Results: P-Value = <2e-16

TukeyHSD Results: (pictured)

	Comparison <chr>	Difference <dbl>	P-Value <dbl>
1	arts,-agriculture,	-534.36386	1.988842e-11
2	construction-agriculture,	-308.72767	3.184097e-11
3	professional,-agriculture,	-567.17088	1.988842e-11
4	transportation-agriculture,	-203.27155	2.078373e-06
5	wholesale-agriculture,	-578.55827	1.988842e-11
6	construction-arts,	225.63619	1.648614e-07
7	professional,-arts,	-32.80702	9.279420e-01
8	transportation-arts,	331.09232	2.090050e-11
9	wholesale-arts,	-44.19441	7.858827e-01
10	professional,-construction	-258.44321	3.813391e-09
11	transportation-construction	105.45613	3.397902e-02
12	wholesale-construction	-269.83060	1.041134e-09
13	transportation-professional,	363.89934	1.995326e-11
14	wholesale-professional,	-11.38739	9.994212e-01
15	wholesale-transportation	-375.28673	1.992795e-11

Conclusions

Findings:

- Highest death rate among Agricultural & Extractive Industries
- Above average death rate for physical industries such as agriculture, transportation, and construction.
- Statistically significant differences in death rate among different industries
- No standout for sedentary industries as expected from prior research

Limitations:

- API categorization prone to errors due to inconsistent variables
- Correlation =/= Causation, despite the high correlation we cannot make any conclusions with this data
- 13 Industry categories were used, potentially grouping together some occupations with stronger differences

Further Research/Action:

- Assistance towards workers with high physical demand such as Agricultural workers
- Advocating for shorter working days, more breaks, or higher wages to reduce overworking

Thank You!

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